SIAN: a local smart-data workflow for narrative analysis

Working paper for methodological review

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Abstract

SIAN is a local workflow for building traceable narrative corpora from public sources: news, open scientific articles, institutional documents, public blogs, forums and web articles. The goal is not large-scale harvesting for its own sake, but a balanced corpus that can support network analysis, semantic grouping, sentiment summaries and multiobjective concept coverage. The methodology follows the spirit of previous work on AI-assisted analysis of media narratives, complex networks and semantic structures published in Contactos [1], while keeping a human validation stage for actors, frames and cultural interpretation.

1 Methodological position

The system does not treat texts as isolated bags of words. It treats them as an information ecosystem. A narrative may appear first in everyday conversation, then move into news coverage, later into institutional documents and finally into scientific interpretation. These layers have different evidential value and different social functions; therefore they must be collected and analyzed separately before being merged.

2 Source layers

Layer	Typical sources	Function
News	GDELT, RSS, curated URLs	Events, controversies and public framing.
Scientific articles	OpenAlex, Crossref, Redalyc, open PDFs	Specialized evidence and conceptual stabilization.
Institutional sources	Government and public agencies	Regulation, alerts, standards and official language.
Forums and blogs	Public forums, blogs and RSS feeds	Organic public conversation when legally accessible.
Other web sources	Open web documents	Residual layer, never silently mixed with news or science.

3 Pipeline

The reference implementation is the mixed spider. Source-specific runs and full runs must use the same plan; only the scope changes.

$$\begin{aligned}\text{configuration} &\to \text{collection plan} \to \text{candidate records}\\ &\to \text{validation} \to \text{deduplication} \to \text{enrichment}\\ &\to \text{JSON corpus} \to \text{analysis}.\end{aligned}$$

All records must preserve source type, URL, medium, year, query variant, geographic scope, extraction status and error messages. This is required for auditability.

4 Narrative network

After cleaning, the system builds a narrative graph. Nodes may represent unigrams, bigrams, trigrams, actors, frames, events or sources. Edges represent co-occurrence or structural relations. Node and edge weights are normalized counts in $[0, 1]$.

Communities are detected with Louvain or comparable methods. The result is exploratory: it helps identify clusters, silences, repeated frames and semantic shifts, but it does not replace interpretation.

5 Human validation

Human validation is part of the method. The user must be able to remove irrelevant nodes, correct actors, merge equivalent phrases, exclude commercial noise and interpret whether a semantic community has cultural meaning.

6 Optimization

The multiobjective model selects a compact set of concepts that covers narrative units, maximizes selected node relevance and minimizes structural damage caused by removing the selected nodes. Feasibility is handled consistently: feasible solutions dominate infeasible ones; among infeasible solutions, the one with smaller coverage violation is preferred.

7 Limits

The system can try to obtain a minimum number of records per source type and year, but it cannot honestly guarantee that such records exist or are legally accessible. It must report coverage gaps instead of hiding them. This is essential for publication.

8 Direct input and batch analysis

SIAN has two input paths. Collection builds a corpus from a search plan. If a JSON or JSONL corpus already exists, *Analyze an existing corpus* opens the analysis views directly. For large corpora, the GitHub Actions workflow runs four stages outside Streamlit: ingestion and audit, lexical analysis, networks and optimization, and descriptive analysis and export. The viewer reads the resulting ZIP. A private URL supplies the corpus, and the remote artifact is encrypted.

Lexical normalization lowercases text, decomposes Unicode with NFKD, removes combining marks, and separates punctuation, hyphens and underscores. It then filters Spanish and English stopwords and merges declared variants, such as *tatuajes* and *tatuaje*. N-grams are consecutive

terms in the retained sequence. Source spelling remains available for quotations. Sentiment analysis retains negators and intensifiers in its own token sequence.

The ZIP contains frequencies and the lexical flower, actors and events, a timeline, frames, idea groups, document and source matrices, sentiment, traceability, evidence and the Smart Data Nucleus. It includes narrative, semantic, knowledge and structural networks. Network figures show a subset of the strongest edges, while tables retain the full networks. Evidence-based relations require human review.

8.1 Cover model

Let x_i indicate selection of node i and r_{ij} indicate the loss of edge (i, j) after removing selected nodes. The model imposes $r_{ij} \geq x_i$, $r_{ij} \geq x_j$ and $r_{ij} \leq x_i + x_j$. It minimizes the number of selected nodes and lost edge weight, and maximizes selected node weight. The three objectives are reported as shares of their respective totals. Linear constraints are stored in sparse matrices. Weighted greedy sweep, MOEA, MOSA and multiobjective MMC use a common evaluation budget and seed. The manifest records the parameters, corpus hash and code revision when the workflow runs on GitHub.

References

[1] Mora-Gutiérrez, R. A. and collaborators. Article in *Contactos*. Available at: https://contactos.izt.uam.mx/index.php/contactos/article/view/625.

9 Update: available evidence and analysis requirements (October 6, 2026)

This update supersedes earlier assumptions of complete metadata availability. Authorship, publication dates and outlets are sought but cannot be guaranteed. Each field preserves its value, evidence, method and state: explicit, inferred, searched but not found, not evaluable, not applicable or conflicting. Conflicting candidates are retained rather than arbitrarily resolved.

Publication, update and consultation dates are distinct. Precision is year, month or day; missing components are never filled. The legacy search year is preserved as `search_year`; `year` only represents usable explicit publication evidence. Index observation dates are not publication dates. A PDF link alone is not evidence of full-text retrieval.

Documents without authors or dates remain available for thematic exploration. Each analysis reports eligible documents, corpus total, exclusion reasons and coverage by source type. Coverage does not establish representativeness or statistical sufficiency. Legacy ambiguous dates require human review for temporal comparison. A URL domain is not an author, identified publisher or event location.

9.1 Descriptive models and review

The document–concept matrix uses 1 for reviewed presence, 0 only for absence after explicit concept review, and missing values for unreviewed cases. Source–concept proportions use concept-specific reviewed denominators. Actor–concept–stance counts keep support, rejection and ambivalence separate, requiring a verbatim quotation and reviewed speaker attribution. Directed relationships retain their type, document and supporting quotation. Causation asserted in a text is not an estimated causal effect. Temporal counts cover the intersection of reviewed and explicitly dated documents, with coverage reported. They do not estimate population attitudes.

Existing heuristic modules produce exploratory candidates. `evidence_model.py` checks metadata states and quotations, not disciplinary truth. JSON reviews are preserved through corpus export. Reception must be supported by response evidence, not inferred from a message alone. Documents can depend on the same institution or event; inferential testing and diffusion estimation are not implemented by these descriptive counts.

9.2 Pseudocode

1. Retrieve content and provenance; search labelled metadata when possible.
2. Preserve field values, evidence, states, conflicts and date precision.
3. Separate publication, update, consultation and search year.
4. Check requirements per analysis; record inclusion or exclusion reason.
5. Check verbatim quotations and attribution in submitted reviews.
6. Build matrices with missing values distinct from zero; preserve stance categories.
7. Export coverage, matrices, relations and limitations of the used subset.

The complete policy and reproducible commands are in `EVIDENCIA_Y_MODELOS.md`. Updating the software does not turn historical reports into new measurements.

9.3 Shared corpus contract

Identity and evidence-preserving merging are centralized in `corpus_contract.py`, shared by the interface and command line tools. Identifier URL parameters, retrieval provenance, text variants and conflicts are retained. Similar titles do not establish document identity. Lexical non-detection in recovered fragments does not prove intentional silence. Annual dates do not establish within-year order. Relational exports retain empty reviewed-model tables when evidence is insufficient. The integral audit documents pilot limits.

9.4 Versioned records and stages

Document, version and retrieval identities are separate. Reviewed selection requires rationale, evidence and reviewer; changes invalidate prior reviews. Titles, abstracts and fragments remain distinct. Six stages retain integrity checked checkpoints for resumption. Saved-corpus processing organizes existing evidence rather than simulating new retrievals. Query caching, bounded retries, transactional storage and atomic writes support the versioned corpus contract.